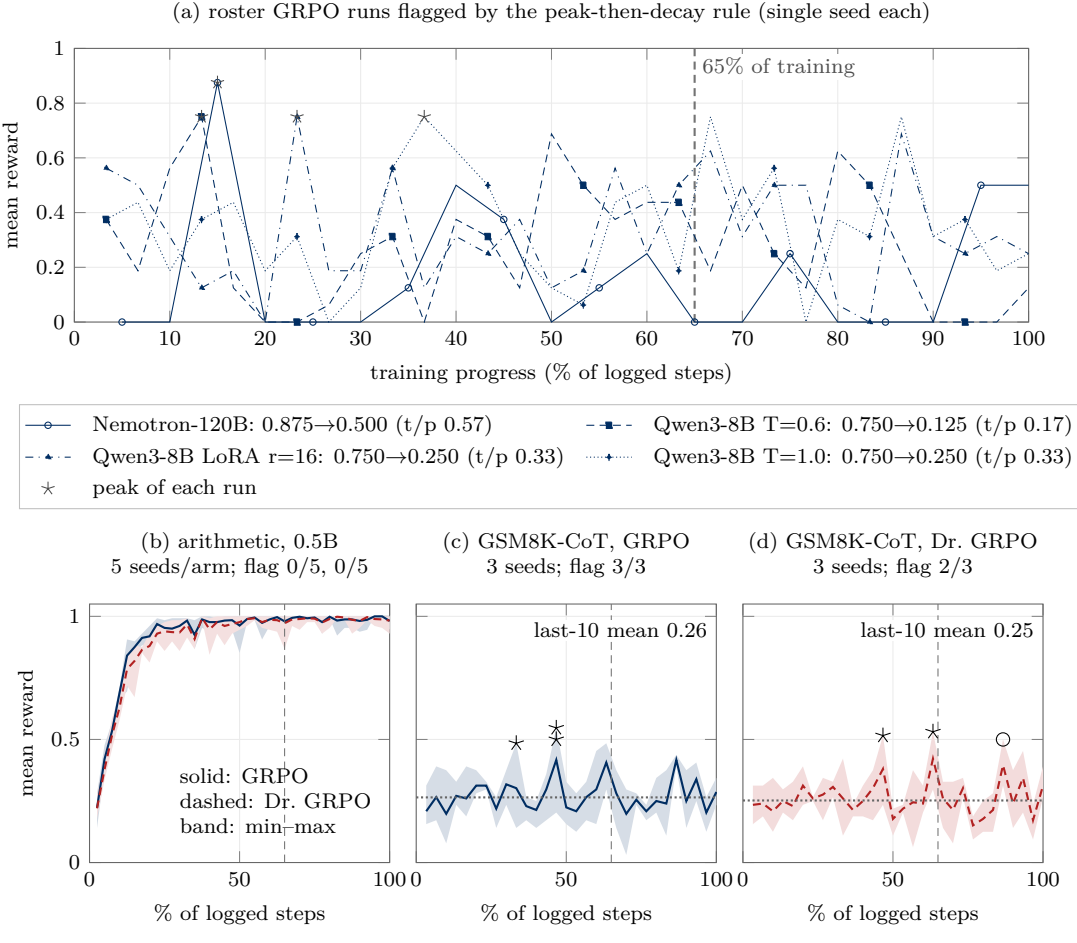
(a) roster GRPO runs flagged by the peak-then-decay rule (single seed each)
mean reward
training progress (% of logged steps)
65% of training
Nemotron-120B: 0.875→0.500 (t/p 0.57)
Qwen3-8B T=0.6: 0.750→0.125 (t/p 0.17)
Qwen3-8B LoRA r=16: 0.750→0.250 (t/p 0.33)
Qwen3-8B T=1.0: 0.750→0.250 (t/p 0.33)
peak of each run
(b) arithmetic, 0.5B
5 seeds/arm; flag 0/5, 0/5
(c) GSM8K-CoT, GRPO
3 seeds; flag 3/3
(d) GSM8K-CoT, Dr. GRPO
3 seeds; flag 2/3
mean reward
solid: GRPO
dashed: Dr. GRPO
band: min–max
last-10 mean 0.26
last-10 mean 0.25
% of logged steps
% of logged steps
% of logged steps